Unit 6 — Functions & *Linear Relationships*

Name _____ Date _____ Period _____

Fill in the blanks as we go. For each practice question, show your work in the space given and box your answer.

🎯 By the end of this unit, I can…

- Decide whether a relation is a function
- Find and interpret slope and y-intercept in context
- Compare functions shown as graphs, tables, and equations
- Solve systems of linear equations

6.1 Functions

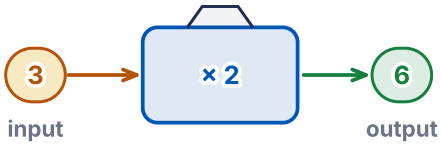
NC.8.F.1 — A FUNCTION IS A RULE THAT GIVES EACH INPUT EXACTLY ONE OUTPUT.

What is this section about? A function is like a vending machine. You press one button (the input). You get one snack (the output). If one button ever gave you two different snacks, the machine would be broken. That broken machine is *not* a function.

The Rule

A function gives each input exactly _____ output.

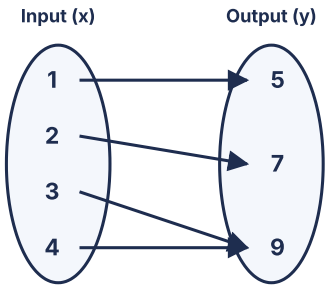
The input is usually _____. The output is usually _____.



Put 3 in. The rule "× 2" sends out 6.

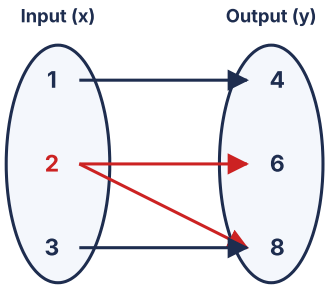
💡 **Outputs may repeat.** Two inputs can share the same output. Only an *input* with two different outputs breaks the rule.

See It: Mapping Diagrams



Function ✓

Each input has one arrow. 3 and 4 share the output 9. That is fine.

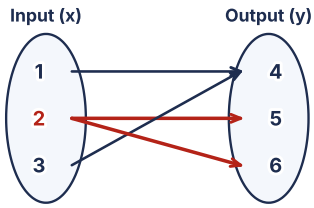


Not a function ×

The input 2 has two arrows: one to 6 and one to 8.

EXAMPLE

Worked example: Is this a function? (1, 4), (2, 5), (3, 4), (2, 6)

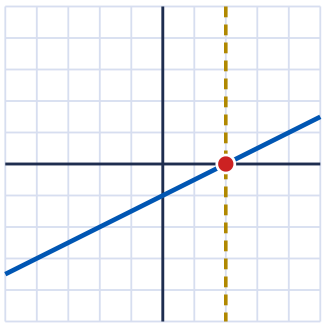


The input 2 has two arrows (to 5 and to 6).

Step 1. List the inputs (the first numbers): 1, 2, 3, 2.
Step 2. Look for an input that shows up twice. The input 2 shows up twice.
Step 3. Check its outputs. (2, 5) and (2, 6): the input 2 has two different outputs.
Step 4. So this is **not** a function.

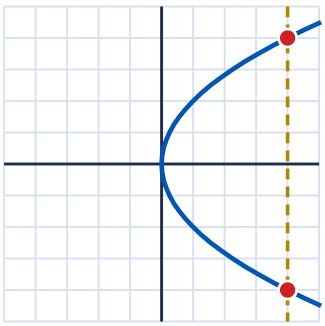
The Vertical Line Test (for graphs)

Slide a vertical line across the graph. If it ever touches the graph more than _____, the graph is not a function.



Function ✓

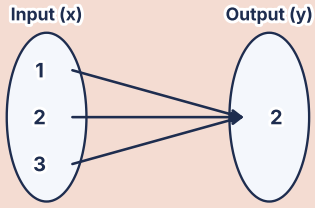
Every vertical line touches it once.



Not a function ×

The line at x = 4 touches it twice.

Find the mistake. Jay says (1, 2), (2, 2), (3, 2) is *not* a function, because the 2 shows up three times. Is Jay right? Write yes or no: _____

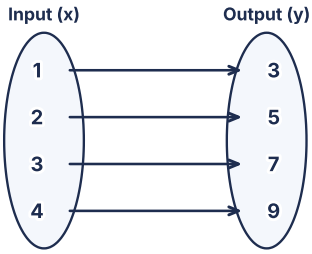


Jay's points as a mapping diagram. Count the arrows leaving each input.

TRY IT

Practice 1: Is this table a function? Write yes or no.

x	1	2	3	4
y	3	5	7	9

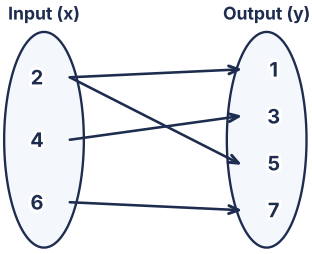


The table as a mapping diagram

Function? _____

TRY IT

Practice 2: (2, 1), (4, 3), (2, 5), (6, 7). Function? Write yes or no.

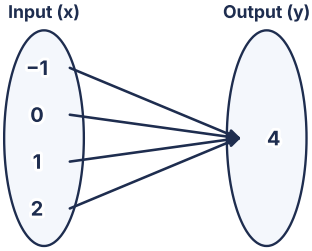


Count the arrows leaving each input.

Function? _____

TRY IT

Practice 3: (−1, 4), (0, 4), (1, 4), (2, 4). Function? Write yes or no.

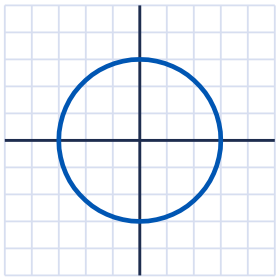


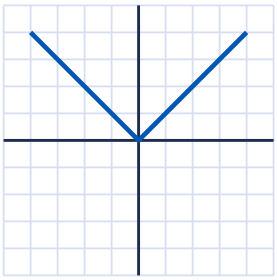
Count the arrows leaving each input.

Function? _____

TRY IT

Practice 4 and 5: Use the vertical line test. Write yes or no for each graph.





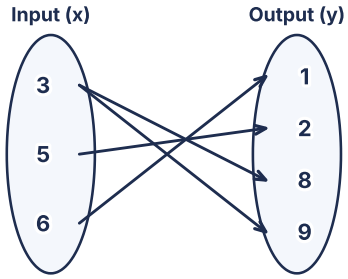
Circle: function? _____

V shape: function? _____

✏ Practice: Functions

1. ★ Try it Is this a function? (3, 8), (5, 2), (3, 9), (6, 1)

- ○ Yes
- ○ No



Count the arrows leaving each input.

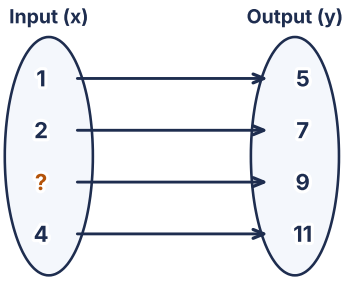
2. ★★ On level This table is **not** a function. Which input breaks the rule? Write just the number.

x	2	4	6	4
y	1	3	5	7

Answer: _____

3. ★★★ Challenge (1, 5), (2, 7), (☐, 9), (4, 11). Give one number for the box that makes this **NOT** a function.

Answer: _____



Which input could the ? be, so one input gets two arrows?

6.4 Linear Functions in Real-World Context

NC.8.F.4 — CONSTRUCT A FUNCTION TO MODEL A LINEAR RELATIONSHIP; INTERPRET RATE OF CHANGE AND INITIAL VALUE IN CONTEXT.

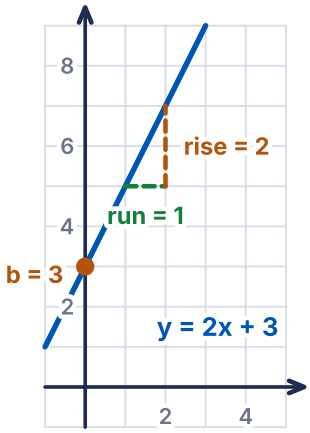
What is this section about? Real-world problems often follow a linear pattern — a steady rate of change plus a starting value. We can write equations in y = mx + b form to model these situations and use them to make predictions.

Slope & Y-Intercept in Context

In real-world problems, the slope and y-intercept have specific meanings:

m (slope) = the _____ of change — how much y changes per unit of x.

b (y-intercept) = the _____ value — what y equals when x = _____ .



b is where the line crosses the y-axis. The triangle shows m: rise ÷ run = 2 ÷ 1 = 2.

💡 **Quick check:** The y-intercept is always the "before-anything-happens" value — the initial cost, starting amount, or fixed fee.

Keywords for Slope and Y-Intercept

Slope keywords (rate):

- per hour, per mile, per day
- each, every
- increases by, decreases by
- price/cost per item

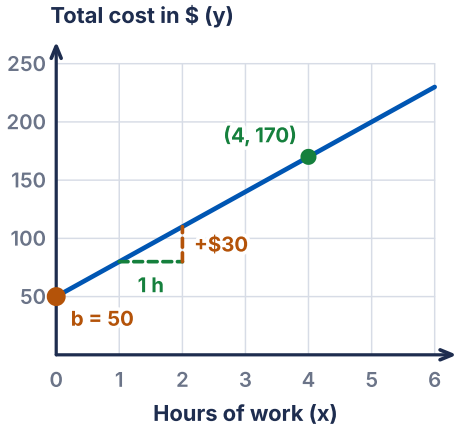
Y-intercept keywords (start):

- flat fee, base cost
- starting amount, initial
- up-front, deposit
- "to begin with"

EXAMPLE

Plumber example: A plumber charges $50 to come to your house, plus $30 per hour of work.

```
The flat fee ($50) is paid before any work happens → b = 50
The $30 per hour is the rate of change → m = 30
Equation: y = 30x + 50
```

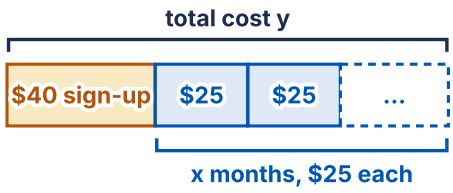


Start at $50 on the y-axis, then go up $30 for each hour.

Interpret: If x = 4 hours, then y = 30(4) + 50 = $170 total.

TRY IT

Practice 1: A gym charges a $40 sign-up fee plus $25 per month. Write the equation in slope-intercept form for the total cost *y* after *x* months.



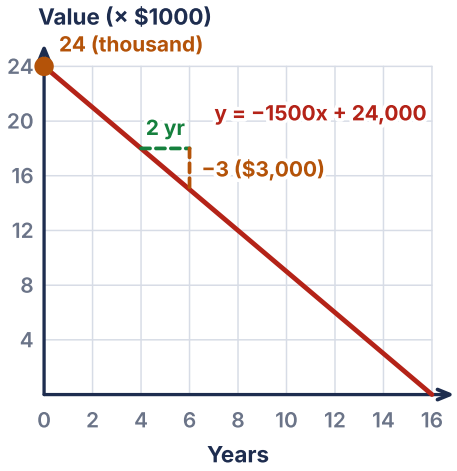
Which part is paid once? Which part repeats?

Slope (m) = _____ · Y-intercept (b) = _____

Equation: y = _____ x + _____

EXAMPLE

Car value example: The value of a car after x years is modeled by $y = -1500x + 24{,}000$.

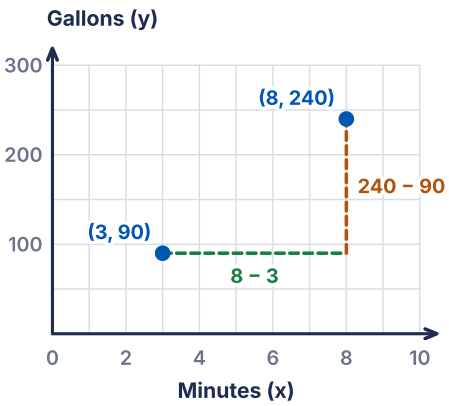


The line starts at 24 on the y-axis ($24,000 when new) and goes down $3,000 every 2 years ($1,500 per year).

`m = −1500 → the car loses` **`$1,500 per year`** `(slope is negative because value decreases)`
`b = 24,000 → the car was worth` **`$24,000`** `when new (when x = 0)`

TRY IT

Practice 2: A swimming pool is being filled. After 3 minutes there are 90 gallons; after 8 minutes there are 240 gallons. The pool was empty before filling started.



Rise ÷ run gives gallons per minute.

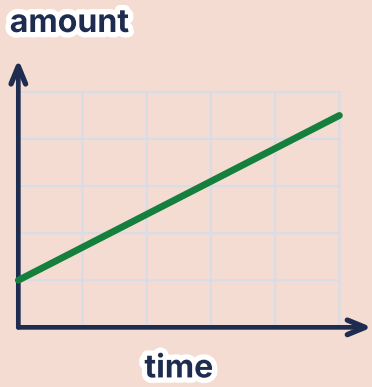
Find the slope (gallons per minute): m = _____

The y-intercept (initial gallons) = _____

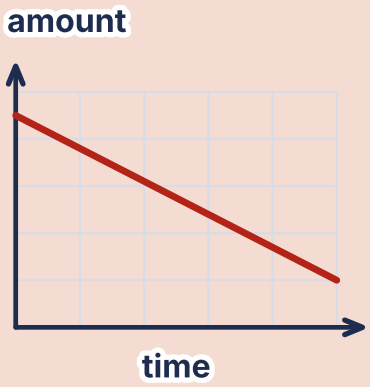
Equation: y = _____ x

Hint: slope = (240 − 90) ÷ (8 − 3) = 150 ÷ 5 = ?

⚠️ **Watch the sign of the slope:** If the quantity is *increasing* over time (filling, growing, earning), m is **positive**. If *decreasing* (depreciating, draining, using up), m is **negative**.



Filling: goes up, m > 0



Draining: goes down, m < 0

TRY IT

Practice 3: Your phone has 80% battery and loses 5% per hour of use. Write an equation for battery percent y after x hours.



Start at 80%. Lose 5% every hour.

m = _____ · b = _____

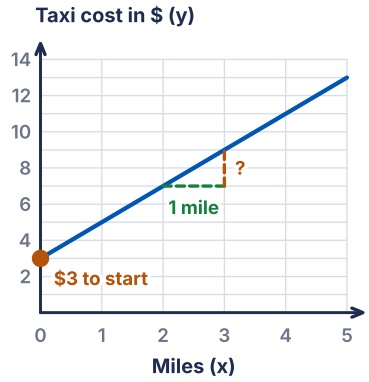
Equation: y = _____ x + _____

After 4 hours, battery = _____ %

✏ Practice: Linear Functions in Real-World Context

1. ★ Try it A taxi charges $3 to start, plus $2 per mile. What is the rate of change (slope)? Write just the number.

Answer: _____



Each mile, how much does the cost go up?

2. ★★ On level Bowling costs $4 for shoes, plus $5 per game. Write an equation for the total cost y after x games.

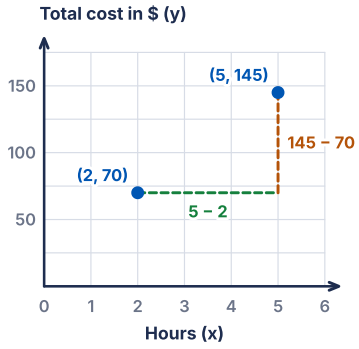
Answer: _____



Paid once: shoes. Repeats: each game.

3. ★★★ Challenge A tutor charges a fixed fee plus a rate per hour. 2 hours cost $70. 5 hours cost $145. What is the fixed fee? Write just the number.

Answer: _____



The fixed fee is the cost at 0 hours (on the y-axis).

6.5 Comparing Functions

NC.8.F.2 — COMPARE PROPERTIES OF TWO FUNCTIONS EACH REPRESENTED IN A DIFFERENT WAY (ALGEBRAICALLY, GRAPHICALLY, IN A TABLE, OR VERBALLY).

The big idea: Linear functions can be shown four ways — equation, table, graph, or verbal description. To compare two functions, get both into y = mx + b form (or just identify their slopes and y-intercepts). Then you can compare directly.

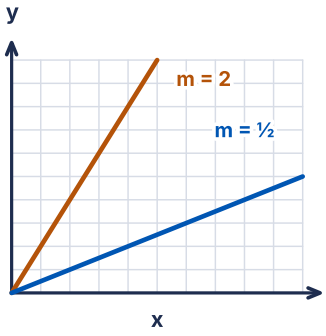
The Strategy

Before answering *any* comparison question, find the slope (m) and y-intercept (b) for **BOTH** functions first.

To compare *rate of change*, look at: _____

To compare *starting value*, look at: _____

The function with the steeper graph has the _____ slope.



Two lines from the same start. Compare how fast each one climbs.

EXAMPLE

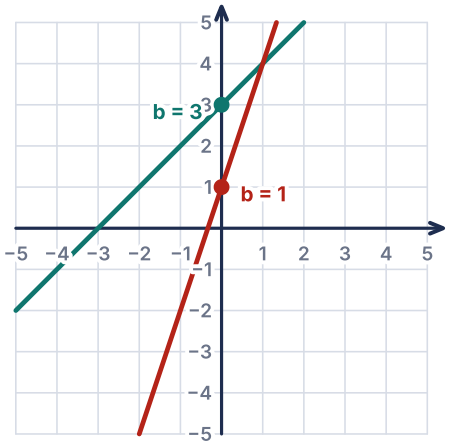
Compare these two functions:

Function A (equation):

y = x + 3

Function B (table):

x	0	1	2	3
y	1	4	7	10



A: y = x + 3 · B: y = 3x + 1 (steeper)

```
Function A: m = 1, b = 3
Function B: m = 3 (each row, y goes up by 3); b = 1 (when x = 0)
```

Greater rate of change? Function B (slope of 3 > slope of 1).

Higher starting value? Function A (b = 3 > b = 1).

TRY IT

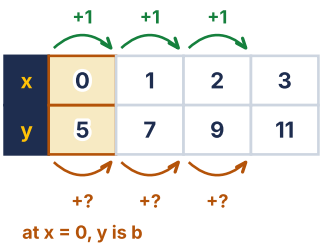
Practice 1: Compare these two functions:

Function A:

y = 4x + 2

Function B:

x	0	1	2	3
y	5	7	9	11



Function B: how much does y change each time x goes up 1?

Function A: m = _____, b = _____

Function B: m = _____, b = _____

Which has the higher rate of change? Function _____

Which has the higher starting value? Function _____

EXAMPLE

Cell phone plans: Plan A costs $20/month plus $0.05/minute. Plan B costs $30/month plus $0.03/minute.

```
Plan A: y = 0.05x + 20
Plan B: y = 0.03x + 30
At x = 0 minutes: Plan A = $20, Plan B = $30 → Plan A starts cheaper
At x = 500 minutes: Plan A = $45, Plan B = $45 → tied!
At x = 1000 minutes: Plan A = $70, Plan B = $60 → Plan B is cheaper for heavy use
```



Plan A starts lower but climbs faster than Plan B.

TRY IT

Practice 2: Two savings accounts:

- Account A starts with $300 and adds $25/week → y = 25x + 300
- Account B starts with $400 and adds $20/week → y = 20x + 400

Which has more money after 10 weeks?



Same 10 weeks, different start and different rate.

Account A after 10 weeks: y = 25(_____) + 300 = $_____

Account B after 10 weeks: y = 20(10) + 400 = $_____

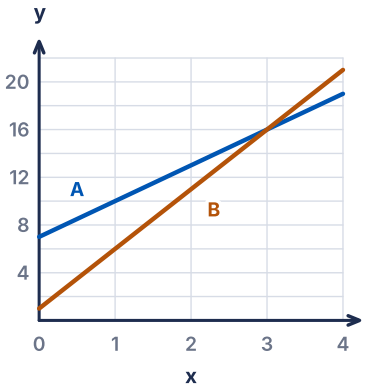
Account _____ has more after 10 weeks.

🔑 **Strategy:** When word problems compare two situations, write *both* equations first, then plug in the same value of x to compare. Or, if asked about long-term, look at the slopes — steeper slope wins eventually.

✏ Practice: Comparing Functions

1. ★ Try it Function A: y = 3x + 7. Function B: y = 5x + 1. Which has the greater rate of change?

- ○ Function A
- ○ Function B



A: y = 3x + 7 · B: y = 5x + 1. Which line is steeper?

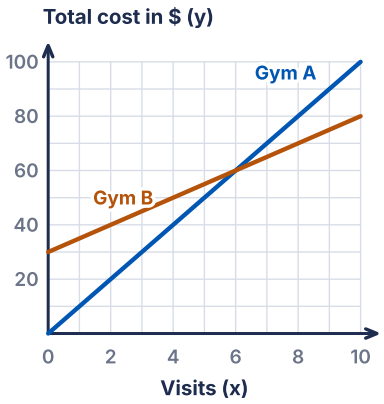
2. ★★ On level Function A is the table. Function B is y = 4x + 9. Which has the greater rate of change?

x	0	1	2	3
y	4	10	16	22

- ○ Function A
- ○ Function B
- ○ They are the same

3. ★★★ Challenge Gym A costs $10 per visit. Gym B costs a $30 fee plus $5 per visit. After how many visits do they cost the same? Write just the number.

Visits	Gym A	Gym B
0	$0	$30
1	$10	$35
2	$20	$40
3	…	…



Answer: _____

6.6 Qualitative Graphs

NC.8.F.5 — DESCRIBE QUALITATIVELY THE FUNCTIONAL RELATIONSHIP BETWEEN TWO QUANTITIES BY ANALYZING A GRAPH; SKETCH A GRAPH FROM A VERBAL DESCRIPTION.

What's a qualitative graph? A graph that shows the *shape* of a relationship over time without exact numbers. The story of the graph matters more than specific values. Common examples: distance-time graphs, temperature-time graphs.

Reading the Shape

Each section of a qualitative graph tells you something. Going left-to-right, ask:

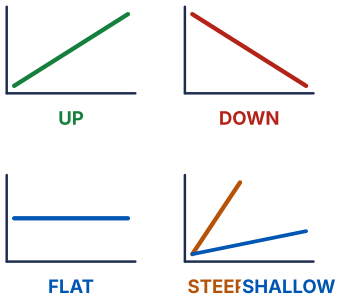
Going UP means the quantity is _____.

Going DOWN means the quantity is _____.

FLAT (horizontal) means the quantity is _____ (not changing).

STEEPER lines mean the change is _____.

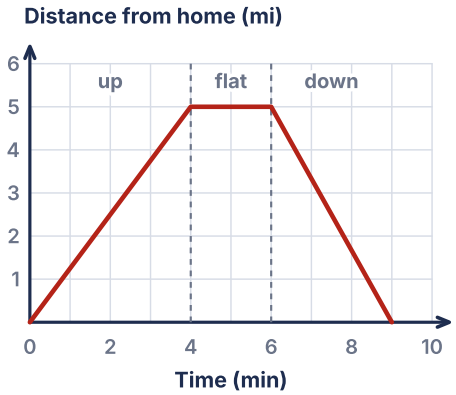
SHALLOWER (gentler) lines mean the change is _____.



The four shapes to look for in any graph.

EXAMPLE

Maria rides her bike to school:



Maria's Bike Ride

The graph shows Maria's distance from home over time. We can read the story:

- **0 to 4 min:** line goes UP at a steady rate → Maria is biking *away* from home toward school.
- **4 to 6 min:** line is FLAT → Maria stays at school (distance doesn't change).
- **6 to 9 min:** line goes DOWN steeper than it went up → Maria rides home *faster* than she went.

⚠️ **Common mistake:** When someone is *returning home*, the distance-from-home line goes DOWN (toward zero), not back up! The y-axis is "distance from start" — if you're getting closer to home, your distance is decreasing.

TRY IT

Practice 1: A swimmer dives into a pool, swims at a steady pace to the far wall, rests for a moment, then swims back *faster* than she went.

The graph below should show distance from start versus time. Match each phase to the correct description:



The swimmer's distance from the start

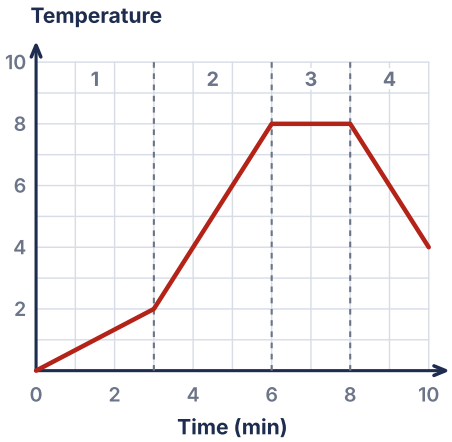
Phase 1 (going up at steady rate): she swam _____ to the far wall at a steady pace.

Phase 2 (flat line): she _____ at the far wall.

Phase 3 (going down steeper): she swam _____ faster than she went.

EXAMPLE

A pot of water: A pot of water is placed on a stove. It heats slowly at first, then more rapidly, stays at boiling for several minutes, then is removed and cools gradually.



Phases 1 to 4 of the pot of water

- **Phase 1 (shallow up):** heats slowly
- **Phase 2 (steep up):** heats more rapidly
- **Phase 3 (flat):** at boiling, temperature stays constant
- **Phase 4 (down):** removed from stove, cools off gradually

TRY IT

Practice 2: Look at a distance-time graph with three phases:

- Phase 1 (0–3 min): steep upward line
- Phase 2 (3–6 min): flat line
- Phase 3 (6–10 min): gradual upward line



The three phases, drawn

Phase 1: the person is moving _____ (away from start).

Phase 2: the person is _____ (not moving).

Phase 3: the person is moving _____ (still away from start).

📐 **Steps for sketching from a story:**

1. Break the story into 2–4 phases.
2. For each phase, decide: increase, decrease, or stay the same?
3. Decide if the change is fast (steep) or slow (shallow).
4. Connect phases at their endpoints — no jumps in the line!

✏ Practice: Qualitative Graphs

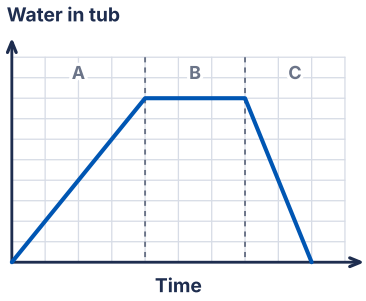
1. ★ Try it A graph shows Leo's distance from home. The line goes DOWN. What is Leo doing?

- ○ Moving away from home
- ○ Moving toward home
- ○ Standing still

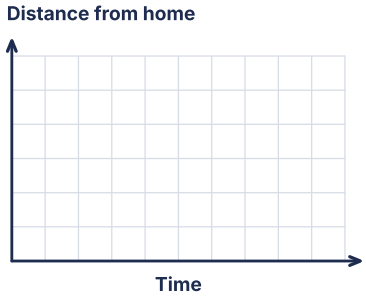


2. ★★ On level This graph shows the water in a bathtub. In which part is the tub draining?

- ○ Part A
- ○ Part B
- ○ Part C



3. ★★★ Challenge Sketch it: Sam walks slowly to the park. He plays there for a while. Then he runs home. Sketch his distance from home over time.



Answer: _____

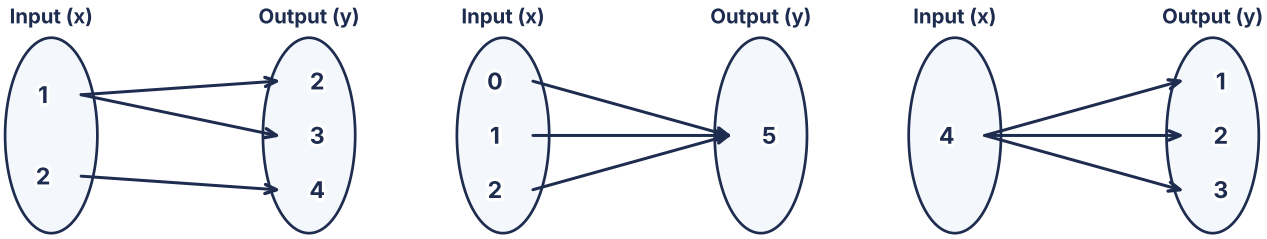
Mixed Practice: All Sections

NC.8.F.1 · NC.8.F.2 · NC.8.F.4 · NC.8.F.5 — REVIEW OF 6.1, 6.4, 6.5 AND 6.6

Put it all together. These questions mix every section. Start with the ★ ones if you want a warm-up. Stuck? Look back at the section's examples. You've got this!

✏ Comprehensive Practice All sections

1. ★ Try it (6.1) Which set of points is a function?



(1, 2), (1, 3), (2, 4) (0, 5), (1, 5), (2, 5) (4, 1), (4, 2), (4, 3)

- ○ (1, 2), (1, 3), (2, 4)
- ○ (0, 5), (1, 5), (2, 5)
- ○ (4, 1), (4, 2), (4, 3)

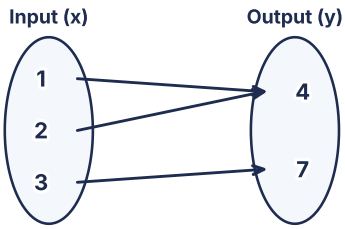
2. ★ Try it (6.1) A function machine has the rule "× 3, then + 2." What comes out when 6 goes in?



Answer: _____

3. ★★ On level (6.1) Is this mapping a function?

- ○ Yes
- ○ No



4. ★ Try it (6.4) A phone plan costs y = 15x + 35, where x is the number of months. What does the 35 mean?

- ○ The cost for each month
- ○ A one-time starting fee
- ○ The number of months



Where is the 35 on the graph? What is x there?

5. ★★ On level (6.4) A candle is 30 cm tall. It burns down 2 cm each hour. Write an equation for its height y after x hours.

Answer: _____

	+1	+1	+1	
x	0	1	2	3
y	30	28	26	24
	?	?	?	

start height

x = hours, y = height in cm. How does y change each hour?

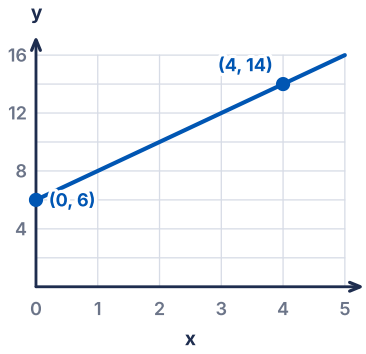
6. ★★★ Challenge (6.4) The table shows the money left on a gift card after x movies. After how many movies will the card have $0? Write just the number.

Movies (x)	0	1	2	3
Money left (y)	$50	$45	$40	$35

Answer: _____

7. ★★ On level (6.5) Function A is the graph. Function B is y = 3x + 1. Which has the greater rate of change?

- ○ Function A
- ○ Function B



8. ★★★ Challenge (6.5) Store A rents bikes for $8 per hour. Store B charges a $12 fee plus $5 per hour. For a 6-hour ride, which store is cheaper?

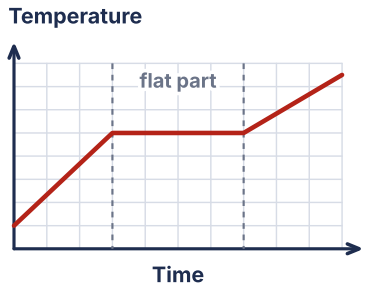
- ○ Store A, by $6
- ○ Store B, by $6
- ○ Store B, by $4
- ○ They cost the same



Which line is lower at 6 hours? Find both costs to see by how much.

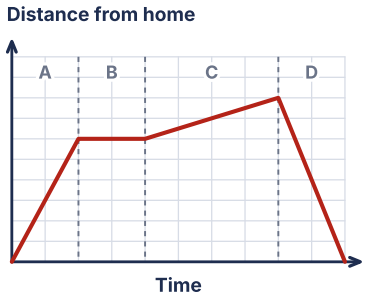
9. ★ Try it (6.6) On a graph of temperature over time, a flat part means the temperature is...

- ○ going up
- ○ going down
- ○ staying the same



10. ★★ On level (6.6) The graph shows Ali's distance from home. In which part is Ali moving, but at his slowest speed?

- ○ Part A
- ○ Part B
- ○ Part C
- ○ Part D



11. ★★★ Challenge (6.6) Find the mistake: Ben walks home from school. He draws his distance from home as a line going UP. What did Ben do wrong?

Answer: _____



Ben's graph